MLFF-GAN: A Multilevel Feature Fusion With GAN for Spatiotemporal Remote Sensing Images

Bingze Song, Peng Liu, Jun Li, *Fellow, IEEE*, Lizhe Wang, *Fellow, IEEE*, Luo Zhang, Guojin He, Lajiao Chen, and Jianbo Liu

***Abstract*— Due to the limitation of technology and budget, it is often difficult for sensors of a single remote sensing satellite to have both high-temporal and high-spatial (HTHS) resolution at the same time. In this article, we proposed a new multilevel feature fusion with generative adversarial network (MLFF-GAN) for generating fusion HTHS images. The MLFF-GAN mainly uses U-net-like architecture, and its generator is composed of three stages: feature extraction, feature fusion, and image reconstruction. In the feature extraction and reconstruction stage, the generator employs the encoding and decoding structure to extract three groups of multilevel features (MLFs), which can cope with the huge difference of resolution between high-resolution images and low-resolution images. In the feature fusion stage, adaptive instance normalization (AdaIN) block is designed to learn the global distribution relationship between multitemporal images, and an attention module (AM) is used to learn the local information weights for the change of small areas. The proposed MLFF-GAN was tested on two Landsat and Moderate-Resolution Imaging Spectroradiometer (MODIS) datasets. Some state-of-the-art algorithms are comprehensively compared with MLFF-GAN. We also carried on the ablation experiment to test the effectiveness of different submodules in the MLFF-GAN. The experiment results and ablation analysis show the better performances of the proposed method when compared with other methods. The code is available at https://github.com/songbingze/MLFF-GAN.**

***Index Terms*— Adaptive instance normalization (AdaIN), generative adversarial network (GAN), spatial attention mechanism, U-net.**

I. Introduction

With the continuous development of Earth observation technology, the demand for remote sensing images from time series is increasing. However, due to technical and budgetary constraints, a single satellite product is often unable to acquire both high-temporal and high-spatial (HTHS) resolution. The satellite with high-spatial resolution usually means a long revisit period. On the contrary, the satellite with a short revisit cycle often only provides low-spatial resolution. It is natural to think that through the combination of multi-platform satellite data, this requirement for the HTHS data can be satisfied. For instance, Landsat-8 and Moderate-Resolution Imaging Spectroradiometer (MODIS) bands have similar spectral response functions. In theory, Landsat images (high-spatial resolution and low-temporal resolution, 30 m, 16 days) and MODIS images (low-spatial resolution and high-temporal resolution, 500 m, 1 day) can be fused to generate HTHS images (30 m, 1 day). The fusion Landsat-like images with land surface details can be used for change detection, land-cover analysis, water monitoring, and so on. In general, spatiotemporal fusion requires n pairs of high-resolution and low-resolution images and m low-resolution images at the prediction times to obtain m high-resolution images. Based on this situation, a large number of spatiotemporal fusion algorithms have been proposed in recent years, which can mainly be divided into three categories, weight-based methods [1]–[5], unmixing-based methods [6]–[12], and learning-based methods [shallow learning [13]–[23] such as Bayesian learning and deep learning [24]–[37] such as convolutional neural network (CNN)].

The weight-based methods assume that the change of land surface reflectance can be predicted by weight summarizing the pixel of temporal–spatial images. The typical weight-based method is spatial and temporal adaptive reflectance fusion model (STARFM) [1] algorithm. STARFM determines the weight by spectrum, distance, and time of the pixels in the sliding window. Enhanced STARFM (ESTARFM) [2] considers the difference between mixed pixels and pure pixels to modify the weights of STARFM. The modified version of ESTARFM (mESTARFM) [3] improves the accuracy of fusion by the new scheme of searching similar pixels. Spatial–Temporal Adaptive Algorithm for mapping Reflectance Change (STAARCH) [4] uses multiple low-resolution images to detect the reflectance changes, so that it can more accurately predict the local spectral changes. Fitting (RM fitting), spatial filtering (SF), and residual compensation (RC), which is abbreviated as Fit-FC [5] uses a regression model to fit and correlate the two observations, then removes the block artifacts by spatial filtering, and finally adopts residual compensation to keep the spectral information.

The advantages of the weight-based methods are as follows: 1) they do not need to learn a large number of parameters from many external data, so they are fast and steady in most cases, and 2) their performances are good in the areas that conform to the homogeneous assumption. The disadvantages are as follows: 1) they assume the homogeneous area in the fusion, which is not enough reasonable in most cases, and 2) their ability of reconstruct details is not strong enough, because its weight model is similar to low-pass filter, which tends to smooth out some details.

Manuscript received August 17, 2021; revised December 23, 2021 and February 11, 2022; accepted March 20, 2022. Date of publication May 24, 2022; date of current version May 31, 2022. This work was supported by the National Natural Science Foundation of China under Grant 61731022 and Grant 41971397. *(Corresponding author: Peng Liu.)*

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Digital Object Identifier 10.1109/TGRS.2022.3169916

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Unmixing-based methods assume that the pixel in low-resolution images is composed of a linear combination of high-resolution end-members. Most of these methods use pixel pre-classification as the end-members. A multisensor multiresolution technique (MMT) [6] classifies high-resolution images into end-members, evaluates the abundance of low-resolution images, and finally reconstructs the images by the assumption of abundance invariant. An spatial–temporal data fusion approach (STDFA) method [7] assumes that land-cover types remain unchanged over time. The reflectance changes of two adjacent high-resolution images obtained by unmixing are assigned to the low-resolution images. On the basis of STDFA, enhanced spatial and temporal data fusion model (ESTDFM) [8] strengthens the classification algorithm and adds sliding window and time weight to enhance the prediction ability of the algorithm. To solve the problem of sensor difference and spatial variability in STDFA, improved STDFA (ISTDFA) [9] uses linear regression to eliminate sensor difference and uses weighted linear mixture model to adjust the spatial variability of surface reflectance. Flexible spatiotemporal data fusion (FSDAF) [10] combines both unmixing and weighting methods. The algorithm calculates the spectral change of the homogeneous area, predicts the spatial change by interpolation algorithm, and finally uses the weighted sum of spectral and spatial features to get the high-resolution image. SFSDAF (an enhanced FSDAF that incorporates subpixel class fraction change information) [11] incorporates subpixel class fraction change information, which makes it more accurate to grasp the change of spectral information. In FSDAF 2.0 [12], through edge detection and change detection, more pure pixels are obtained, which makes the unmixing process more accurate. At the same time, change detection is also applied to generate weights, and then, more accurate predictions are obtained. Unmixing-based methods have very clear physical meaning. It takes fusion as an inverse problem, which is ill-posed. For this kind of method, the abrupt changes and details in high-spatial resolution images can be reconstructed to some extent. However, these methods put more emphasis on the relationship between spectral domain and spatial domain but not temporal domain. In the temporal domain, they usually assume that the invariant is in abundance. This abundance invariant assumption seriously weaken the ability of the method, so that the fusion often shows block effect resulting by the classification for end-members.

Learning-based methods are the type of data-driven methods, using the existing datasets to train the model and find the relationship between spatial and temporal images. Sparse representation or dictionary learning methods are typical shallow learning methods, such as SParse-representation-based SpatioTemporal reflectance Fusion Model (SPSTFM) [13]. Many improvements have been made on this type of methods, such as reducing image references [14], increasing regularization [15], and using structure sparsity [16], [17]. A Bayesian method uses the observation data to solve the spatiotemporal fusion by calculating the maximum *a posteriori* probability. Li *et al.* [18] and Huang *et al.* [19], respectively, adopt covariance function and low-pass filtering to describe the relationship between the input high-spatial resolution image and low-spatial resolution image. In [20]–[22], obtaining information from long time series, they, respectively, adopt the pixel unmixing, joint Gaussian distribution, Kalman filter to target images by Bayesian probability from time-series images. Other studies, such as [23], use Bayesian to select trained dictionaries. Deep learning-based methods often use CNN, which has powerful feature extraction ability with the support of big data [24], [25]. Through the design of network structure and loss function, the adaptability to spatiotemporal fusion can be improved, such as spatio-temporal satellite image fusion using deep CNN (STFDCNN) [26], deep convolutional spatio-temporal fusion network (DCSTFN) [27], enhanced DCSTFN [28], spatio-temporal fusion problem with a two-stream convolutional neural network (StfNet) [29], bias-driven spatio-temporal fusion model (BiaSTF) [31], and spatial, sensor, and temporal spatio-temporal fusion (SSTSTF) [34]. In recent years, generative adversarial network (GAN) methods as a sub-branch of deep learning have achieved great success in image generation [38]–[40], style transfer [41], [42], super-resolution reconstruction [43], [44], and many other fields. GAN methods show many advantages in the field of image generation. In spatiotemporal fusion, GAN methods also already have some applications. Recently, remote sensing image spatio-temporal fusion method using a GAN (STFGAN) [35] follows the idea of super-resolution using a GAN (SRGAN) [43] and solves spatiotemporal fusion as a super-resolution problem. The GAN-based spatio-temporal fusion model (GAN-STFM) [36] designs a more flexible way of image prediction by conditional GAN. CycleGAN-STF [37] takes image fusion as a data augmentation problem and then selects the most informative image as a fusion result. GAN-based methods show very promising performances in spatiotemporal fusion. However, there are many particularities in spatiotemporal fusion that still need to be considered further. The architecture of generator or discriminator, the loss function, and feature extraction or fusion scheme all need to be adapted for spatiotemporal fusion.

For remote sensing spatiotemporal fusion, the difficulties mainly come from three points: 1) in the temporal dimension, the dramatic uncertainty of land feature change (such as flood, phenology, and so on, and land types may also change) is hard to predict by images of adjacent time; 2) in the spatial dimension, the huge difference between high-resolution images and low-resolution images brings great difficulties to reconstruct the detailed textures; and 3) for different sensors, there are inevitably systematic errors in imaging process, such as imaging conditions (atmosphere, or solar zenith angle), device differences (spectral response, or modula transfer function), and so on.

These abovementioned three difficulties make it to be a very challenging problem for spatiotemporal fusion. Most model-driven methods, such as weighted method, unmixing-based method, and so on, only use very few volumes of internal data and very limited data priors, so that they are not competent in comprehensively solving the three difficulty points at the same time. Therefore, it is very necessary to use more of the external datasets to learn more priors and features. With learning-based methods, the features in time, space, and sensor dimension all can be organized in an appropriate and efficient way.

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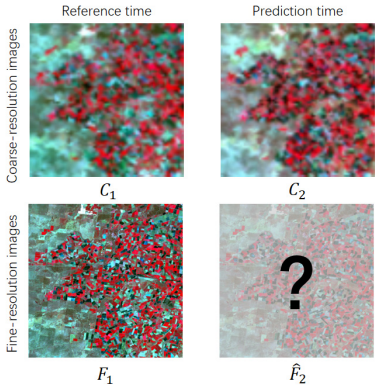


Fig. 1. Objective problem to be solved. Three known images (F_1, C_1, and C_2) are used to obtain the high-resolution image of prediction time ($\hat{F}_2$).

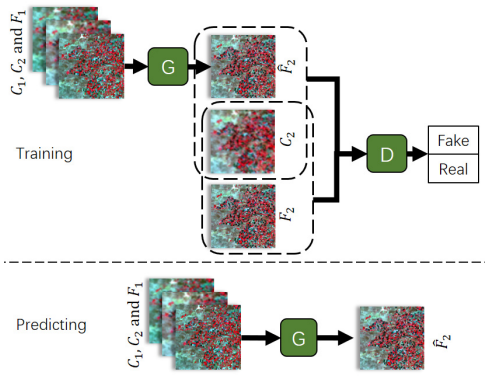


Fig. 2. Overall framework of MLFF-GAN. The generator is responsible for generating the image, and the discriminator is responsible for identifying whether the image is real or fake.

However, the existing data-driven or learning-based methods did not provide very pertinence improvement schemes to the three difficult points of remote sensing spatiotemporal fusion.

In this article, we design an encoding–decoding structure of GAN frame to learn multilevel features (MLFs) for image fusion and specifically and pointedly consider the three difficulty points of spatiotemporal fusion. In the proposed, an MLF fusion GAN (MLFF-GAN) architecture, huge resolution difference, dramatic feature change, and a systematic error of sensor are all considered in a U-net-like generator frame. An attention module (AM) is designed to deal with the changes of image local difference in the time dimension, and adaptive instance normalization (AdaIN) [45] is introduced to learn a systematic error or global difference; at the same time, we construct the backbone structure of the generation network as a U-net-like CNN to deal with huge resolution difference. This comprehensive structure improved the ability of feature learning and generalization ability of MLFF-GAN fusion. We also carry out contrast experiments and ablation experiments to validate the effectiveness of the different parts in MLFF-GAN fusion.

The remaining contents are organized as follows. In Section II, the basic architecture and loss function for MLFF-GAN are presented in detail. In Section III, contrast experiments and ablation experiments are carried out to validate the proposed method. In Section IV, we conclude this article.

II. Methodology

As shown in Fig. 1, we predict the high-resolution image F_2 at time t_2 by a pair of images $\langle F_1, C_1\rangle$ at time t_1 and a low-resolution image C_2 at time t_2. Generally speaking, in the prediction phase, t_1 and t_2 are two close but different times. Similar to other GAN models, as in Fig. 2, MLFF-GAN is composed of two parts: generator and discriminator. The generator is responsible for generating the fusion images through the input data $\langle F_1, C_1\rangle$ and C_2, while the discriminator makes a judgment on the generated image and real image. Generator and discriminator are trained alternately to improve their abilities. After the training, we use generator to predict the fusion image, so the generator is the key point in the proposed method. The generator is designed as a U-net-like structure with three stages: feature extraction, feature fusion, and image reconstruction. The discriminator is a simple CNN structure. In the following, the proposed MLLF-GAN will be addressed in detail.

A. Generative Adversarial Networks

Before illustrating the proposed MLFF-GAN, we first need to simply review the characteristic of traditional GAN. GAN was first proposed in [38], which is composed of generator G and discriminator D. The generator G learns the complex mapping relationship from noise $\boldsymbol{z}$ to real data $\boldsymbol{x}$ and attempts to map the distribution of noise $p_z(\boldsymbol{z})$ to the distribution of real data $p_{\text{data}}(\boldsymbol{x})$. The discriminator D discriminates the difference between the generated data $G(\boldsymbol{z})$ and the real data $\boldsymbol{x}$ through training and outputs a matrix of 0–1 range to judge whether the data are real or fake.

For the spatiotemporal fusion problem, due to the particularity of the input information and ensure the stability of GAN training, the GAN loss of MLFF-GAN adopted is least square GAN (LSGAN). The object function of LSGAN is formulated as

$$\min_D \mathcal{L}_{\text{LSGAN}}(D) = \frac{1}{2}\mathbf{E}_{\boldsymbol{x}}[(D(\boldsymbol{x})-b)^2] + \frac{1}{2}\mathbf{E}_{\boldsymbol{z}}[(D(G(\boldsymbol{z}))-a)^2] \quad (1)$$

$$\min_G \mathcal{L}_{\text{LSGAN}}(G) = \frac{1}{2}\mathbf{E}_{\boldsymbol{z}}[(D(G(\boldsymbol{z}))-c)^2] \quad (2)$$

where a, b, and c are constants. When $b-c=1$ and $b-a=2$, the object function is equivalent to Pearson $\mathcal{X}^2$ divergence.

In the proposed MLFF-GAN, we use conditional LSGAN's object function as its basic GAN framework. The inputs of generator are F_1, C_1, and C_2. At the same time, discriminator is used to ensure the quality of the output images. Besides, MLFF-GAN is constrained not only by the discriminator but also by the combination of different loss functions. The rest of this section will explain the design of MLFF-GAN for its generator, discriminator, and loss function.

B. Generator

The generator of MLLF-GAN consists of three stages: feature extraction, feature fusion, and image reconstruction. The overall architecture of the generator is shown in Fig. 3. We can understand the three stages as a new U-net framework, who is composed of a contracting path (left-hand side, feature extraction), feature fusion (middle part), and an expansive

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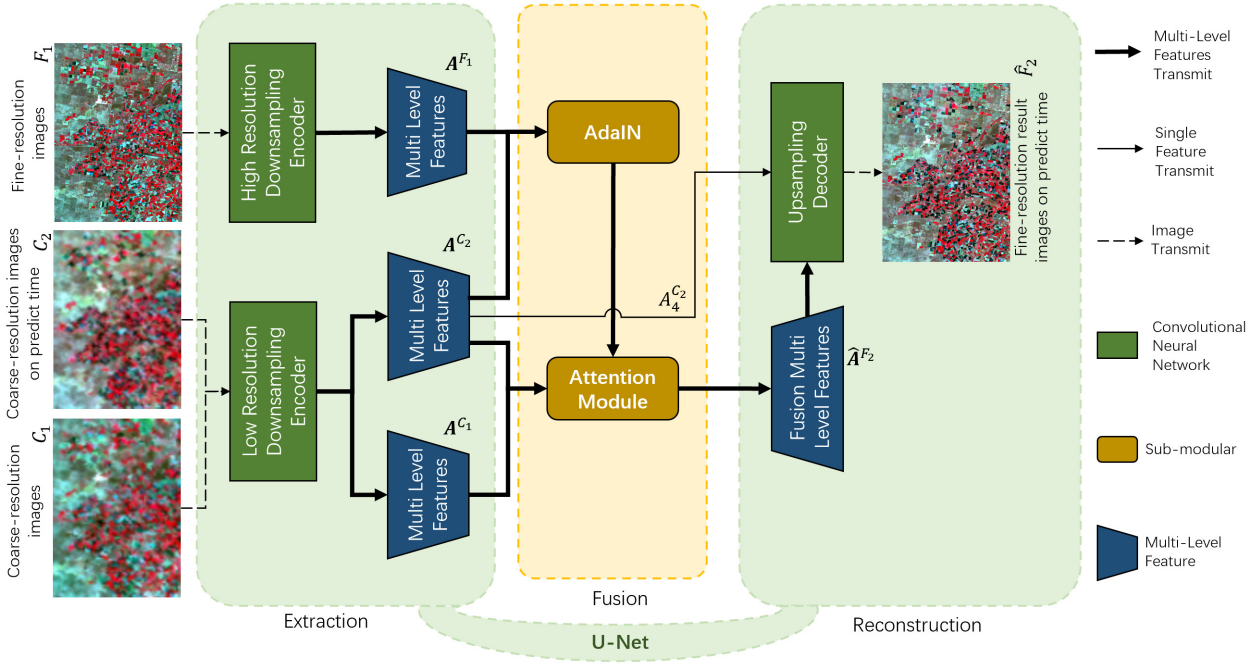


Fig. 3. Generator of MLFF-GAN. The left green rounded rectangle represents the feature extraction stage, the middle orange rounded rectangle represents the feature fusion stage, and the left red rounded rectangle represents the image reconstruction stage. Feature extraction and image reconstruction are symmetrical structures such as U-net. AdaIN is designed for global information transformation, and AM is designed for local change weight learning.

path (right-hand side, reconstruction). The encoding–decoding (extraction and reconstruction) structures are responsible for resolution enhancement in spatial dimensions. The feature fusion part is divided into two modules: global information transformation and local change weight learning. The global information transformation module mainly deals with global distribution and systematic errors by AdaIN. The local change weight module mainly deals with local feature changes in the time dimension by AM. These modules and the encoding–decoding structure can be taken as solutions that correspond to the aforementioned three challenging problems in spatiotemporal fusion.

In the feature extraction stage, multiple downsampling and convolution blocks are connected to extract spatial–temporal features in multisource images. One of the advantages of a deep CNN is that it can extract image features more extensively and deeply than traditional methods. As U-net is a symmetric structure, both the extraction and reconstruction sides use residual block based on residual network (ResNet) [46] as a basic block. Residual block can effectively prevent gradient dispersion and gradient explosion, so that it can smoothly ensure the training of the model. Different from ResNet, the residual block in generator removes batch normalization to obtain a better quality of generated image [47].

As in Fig. 3, three group of features are extracted by the feature extraction stage: one group of spatial features $\mathbf{A}^{F_1}$ and two group of temporal features $\mathbf{A}^{C_1}$ and $\mathbf{A}^{C_2}$, where $\mathbf{A}^{F_1} = \{A_i^{F_1}\}_{i=0}^{N}$, $\mathbf{A}^{F_1} = \{A_i^{F_1}\}_{i=0}^{N}$, $\mathbf{A}^{F_2} = \{A_i^{F_2}\}_{i=0}^{N}$, and N is the number of layers. To obtain the features in different levels, the framework uses the residual downsampling block, that is, to add a convolution layer with a step size of 2 behind the residual block. The number of multilevel layers of the network is determined according to the resolution magnification of a high-resolution image and a low-resolution image. Taking Landsat and MODIS as an example, the ratio of resolution is that $500/30 \approx 2^4$, so the number of downsampling blocks is 4, as shown in Fig. 4(a). With the CNN and downsampling, MLFs are obtained by the feature extraction network. This design decomposes the high-resolution image and exchanges information with the low-resolution image at the same level, so as to provide features for the subsequent fusion stage and reconstruction stage.

In the feature fusion stage, the MLFs of three input images are fused by global information transformation and local change weight learning, as in Fig. 3 (middle part). First, $\mathbf{A}^{F_1}$ and $\mathbf{A}^{C_2}$ are used for global information transformation, which is designed for system errors such as spectral differences or imaging condition differences. Second, $\mathbf{A}^{C_1}$ and $\mathbf{A}^{C_2}$ are used to learn the weight of local change information, which is designed for the abrupt change in land cover or phenological change.

To capture the systematical and global characteristics in the different images, global information transformation uses AdaIN [48] to transform the global distribution of MLFs of C_2 to the MLFs of F_1. AdaIN was originally used for style conversion in GAN. The formula is denoted as

$$\mathrm{AdaIN}(x, y) = \sigma(y)\left(\frac{x - \mu(x)}{\sigma(x)}\right) + \mu(y) \tag{3}$$

where $\mu(x)$ is the mean of x, and $\sigma(x)$ is the variance of x. Through the mean and variance of x and y, AdaIN destyles x and then transfers the style of y into x. In global information transform, x is the MLFs of $\mathbf{A}^{F_1}$, and y is the MLFs of $\mathbf{A}^{C_2}$. Through AdaIN, the overall characteristics of texture and spectrum distribution of F_1 is transformed into the texture and spectrum of C_2; at the same time, the spatial detailed information of each level feature in F_1 is retained. As in Figs. 3 and 4(c), the transformed MLFs are denoted as $\hat{\mathbf{A}}^{F_1} = \{\hat{A}_i^{F_1}\}_{i=0}^{N}$.

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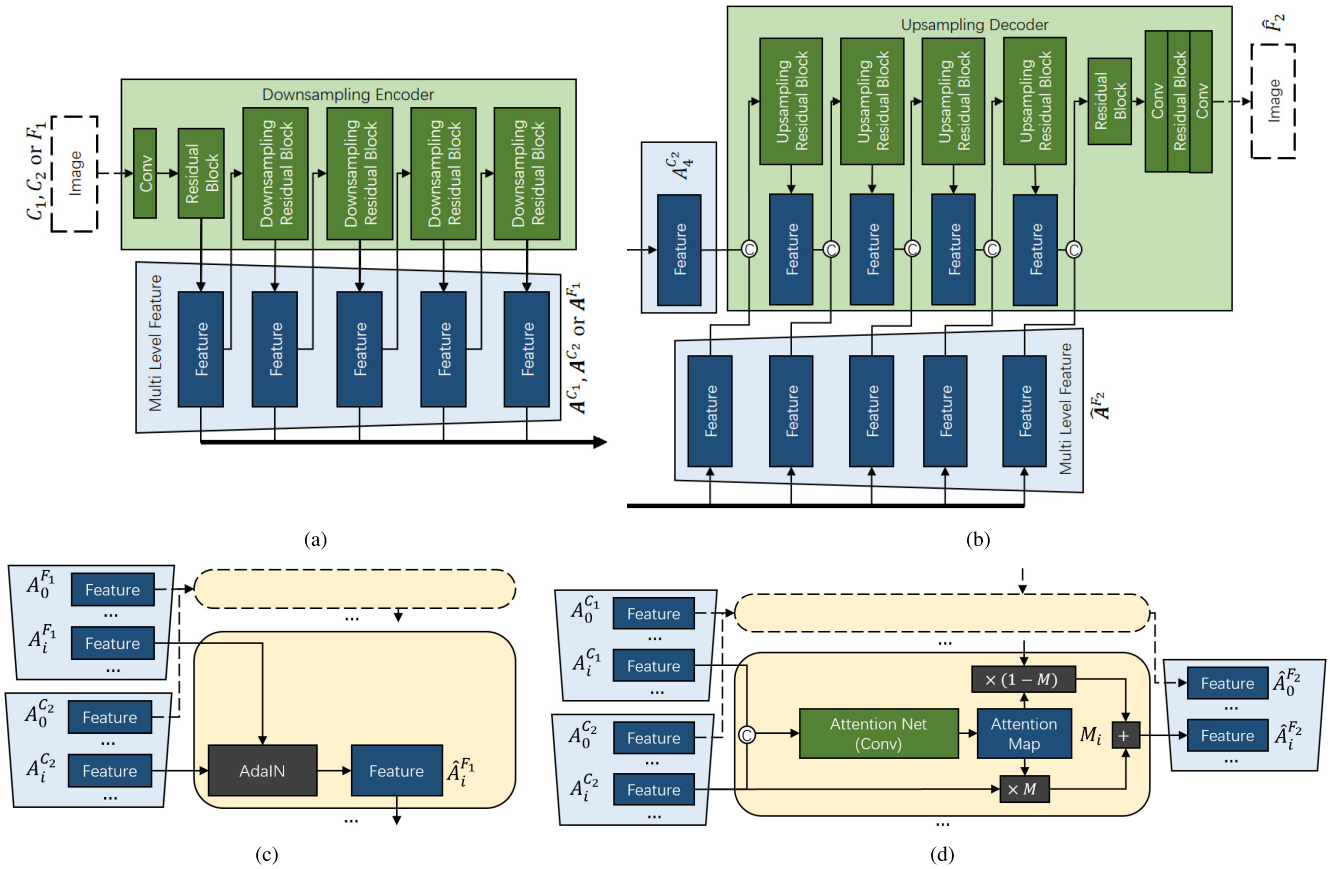


Fig. 4. Detailed structure diagrams for the different parts in Fig. 3. (a) Detailed structure of the feature extraction stage of Fig. 3: the light green rectangle represents the high-resolution downsampling encoder, and the features in the light blue trapezoid are MLFs. (b) Detailed structure in the image reconstruction stage of Fig. 3: circle C is a concatenate operation of two features, the light green rectangle represents the upsampling decoder, and the features in the light blue trapezoid are fusion MLFs. (c) Detailed structure of AdaIN of Fig. 3: the structure in the pale gold rounded rectangle is AdaIN, and the features in the pale blue rectangle represent one of the MLFs. (d) Detailed structure of AM of Fig. 3, where circle C is the same as (b).

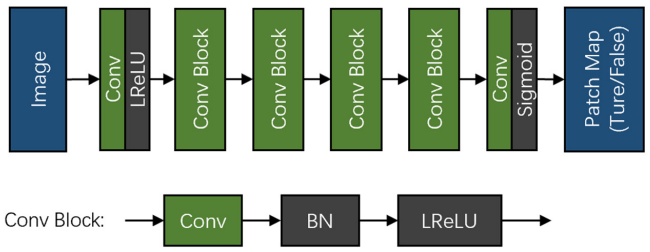


Fig. 5. Discriminator of MLFF-GAN. The discriminator structure is a basic classification neural network, in which convolution block is constructed by convolution, batch normalization, and LeakyReLU activation function. The sigmoid function is used to output the value between 0 and 1.

AM is responsible for learning local changes in the time dimension. As in Figs. 3 and 4(d) (middle part), AM uses the spatial attention mechanism to learn weights from $\mathbf{A}^{C_1}$ and $\mathbf{A}^{C_2}$ to modify $\hat{\mathbf{A}}^{F_1}$ to form $\hat{\mathbf{A}}^{F_2}$. Using the back propagation mechanism of deep learning, the network learns weight maps $\mathbf{M} = \{M_i\}_{i=0}^{N}$ for each level i in the range of 0–1 to assign the salience for the features. The greater the weight, the greater the change, and the greater the proportion will be assigned to $\mathbf{A}^{C_2}$; the smaller the weight, the smaller the change, and the greater the proportion will be assigned to $\hat{\mathbf{A}}^{F_1}$. For each level of $\hat{A}_i^{F_2}$, the formula is defined as

$$\hat{A}_i^{F_2} = M_i A_i^{C_2} + (1 - M_i)\hat{A}_i^{F_1} \tag{4}$$

where M_i is the weight map for the ith layer, in MLFF-GAN, where $i = 0, 1, 2, 3$, and 4. Each level of feature corresponds to a weight map. The weight map is a single-level 0–1 matrix output by a single-layer convolution with one output channel and sigmoid function. The MLF $\hat{\mathbf{A}}^{F_2}$ and the last-level feature $A_4^{C_2}$ of MLF $\mathbf{A}^{C_2}$ are used to guide the upsampling reconstruction network.

For image reconstruction stage, the schematic of reconstruction network is shown in Figs. 3 and 4(b) (right-hand side). After the extracted features, $\mathbf{A}^{F_1}$, $\mathbf{A}^{C_1}$, and $\mathbf{A}^{C_2}$ are integrated into the new fine feature $\hat{\mathbf{A}}^{F_2}$ in the fusion stage. $\hat{\mathbf{A}}^{F_2}$ will be reconstructed into the final fusion image $\hat{F}_2$.

The reconstruction stage is symmetrical to the extraction stage as the U-net architecture. It is composed of convolution and upsampling blocks. In the process of upsampling, there are two groups of input streams. The first group is the last feature from the downsampling encoder feature of the low-spatial resolution image at time t_2 (feature $A_4^{C_2}$), and the second group is the fusion features $\hat{\mathbf{A}}^{F_2}$ generated by AdaIN and AM. The

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two inputs are combined to reconstruct the final image by continuous upsampling. $A_4^{C_2}$ and a series of new features generated in the upsampling decoder are considered to be the features of a high-resolution image at predicted time t_2. Also, $\hat{\mathbf{A}}^{F_2}$ is considered as reference features, which are concatenated with the features generated in the upsampling decoder and input into the upsampling residual block, as shown in Fig. 4(b).

Similar to the extraction network, the reconstruction network also uses residual blocks to ensure the steadiness of gradient descent. The deconvolution layer is added in front of the residual block to form upsampling residual block to double the feature size. The convolution inputs of each block in the reconstruction network are features from different module, and then, the output image will carry on features from different levels by the reconstruction process. There are two advantages in the architectures: 1) for large resolution difference, the spatial matching relationship between downsampling and upsampling can be helpful to the resolution enhancement, and 2) when predicting the features of medium- and high-resolution images at different scales, it is more conducive to learn invariant features and dynamic features in the process of image reconstruction.

C. Discriminator

As shown in Fig. 5, the discriminator of MLFF-GAN is a simple binary CNN, which is basically the same as PatchGAN [41]. Different from the traditional discriminator in [38], in MLLF-GAN, the discriminator is used to judge whether the image patch is a real observation image or a fusion one. Each of the blocks of the discriminator consists of multiple convolution layers, instance normalization layer [49], and leaky rectified linear unit (LeakyReLU) activation function. Convolution operation keeps the size of the receptive field at 70×70. In the training stage of the discriminator, when its inputs are F_2 and C_2, it is expected to get all true matrix, and when its inputs are $\hat{F}_2$ and C_2, it is expected to get all false matrix. This discriminator has far fewer parameters than the generator, because it only needs to keep abstracting and generalizing by deep networks. The adversarial function provided by the discriminator can help the generator to obtain higher quality results. Also, for the weak relative position relationship in a single remote sensing image, the patch identification method has certain advantages.

D. Loss Function

The loss function of the generator in the proposed MLFF-GAN is mainly composed of two types: GAN loss ($\mathcal{L}_{\text{GAN}}$) and the loss for image contents. Image content loss includes L_1 loss, spectrum loss, and structure loss.

The GAN loss is the same as the corresponding part in LSGAN, which is denoted as

$$\mathcal{L}_{\text{GAN}}(G) = \frac{1}{2}\mathbf{E}_{F_1,C_1,C_2}[(D(G(F_1,C_1,C_2),C_2)-1)^2] \quad (5)$$

$$\mathcal{L}_{\text{GAN}}(D) = \frac{1}{2}\mathbf{E}_{F_2,C_2}[(D(F_2,C_2)-1)^2] + \frac{1}{2}\mathbf{E}_{F_1,C_1,C_2}[(D(G(F_1,C_1,C_2),C_2))^2]. \quad (6)$$

$\mathcal{L}_{\text{GAN}}$ is more like a traditional loss function, which can ensure the integral distribution of the output image dataset to be the same as the distribution of the input image dataset.

The purpose of the discriminator is to identify whether the generated image is real or fake, so $\mathcal{L}_{\text{GAN}}(D)$ is the final loss of the discriminator. However, to keep the constraints on the generated image, besides $\mathcal{L}_{\text{GAN}}(G)$, the loss for the image itself is designed to ensure the authenticity of the generated image. We notice that there are many methods [28], [35], [36] that use the perceptual loss calculated by the feature of the self-designed pre-training network to ensure the quality of image generation. Also, in the style transfer and super-resolution [43], [51], [52] for natural images, the features of the pre-trained VGG network are often used to calculate the perceptual loss to obtain better image results. In the experiment, we found that using the perceptual loss designed from the feature of pre-trained VGG makes the overall spectrum and texture of the image change greatly, because the nature image and the satellite image are very different in essence. Also, the image generated using the perceptual loss designed from the feature of self-designed pre-training network is almost the same as the image generated using the loss for image contents, that is, the loss in MLFF-GAN. Even though, the images generated using the loss for image contents make the quantitative evaluation better. Therefore, our loss function is calculated directly on the image, rather than on the features such as the perception loss function.

L_1 loss is used to keep the value of the generated image and the real image at the same pixel position equal, which is denoted as in (7). To obtain the spectral and structural authenticity of the generated remote sensing image, the cosine similarity and the multiscale structural similarity (MS-SSIM) [53] between the real image and the generated image are calculated, respectively. The cosine similarity is denoted as in (8), and the MS-SSIM is denoted as in (9)

$$\mathcal{L}_1 = \frac{1}{K}\sum_{k=1}^{K}\|\boldsymbol{T}-\boldsymbol{P}\|_1 \quad (7)$$

$$\mathcal{L}_{\text{spectrum}} = \boldsymbol{I} - \frac{\boldsymbol{T}\cdot\boldsymbol{P}}{\|\boldsymbol{T}\|_2\|\boldsymbol{P}\|_2} \quad (8)$$

$$\mathcal{L}_{\text{structure}} = \boldsymbol{I} - [l_m(\boldsymbol{T},\boldsymbol{P})]^{\nu_N} \cdot \prod_{j=1}^{N}[c_j(\boldsymbol{T},\boldsymbol{P})]^{\rho_j}[s_j(\boldsymbol{T},\boldsymbol{P})]^{\mu_j} \quad (9)$$

where $\boldsymbol{T}$ is the ground truth image matrix (such as F_2), $\boldsymbol{P}$ is the prediction image matrix [such as $G(F_1,C_1,C_2)$], $\|\cdot\|_p$ is the p-norm, K is the pixels number of image, and $\boldsymbol{I}$ is the tensor of ones. In (9), N is the number of the highest scale, l_j, c_j, and s_j are the luminance, the contrast comparison, and the structure comparison at the jth scale. ν_N, ρ_j, and ν_j are the corresponding weight coefficients.

Finally, the total loss of the generator is

$$\mathcal{L}(G) = \alpha\mathcal{L}_{\text{GAN}}(G) + \beta\mathcal{L}_1 + \gamma\,\mathcal{L}_{\text{spectrum}} + \delta\mathcal{L}_{\text{structure}} \quad (10)$$

where α, β, γ, and δ are the weight coefficients.

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III. Experiments and Results

A. Study Areas and Datasets

In the experiments, the open-source datasets of the Coleambally Irrigation Area (CIA)[1] and the lower Gwydir catchments (LGC)[2] are used to test the proposed model.

CIA is an image dataset of a summer crop growing area in Australia. The height is 43 km in the north–south direction, and the width is 51 km in the east–west direction. The dataset is composed of image pairs of Landsat-7 Enhanced Thematic Mapper Plus (Landsat-7 ETM+) and MODIS MODTRAN4 products. The image size of Landsat-7 ETM+ is 1720×2040. For each pair of images, the MODIS image is interpolated into the same size as the Landsat-7 image. In CIA and LGC, the band numbers of Landsat-7 ETM+ and MODIS images are both six. There are 17 pairs of cloudless images in the CIA dataset. In CIA, the phenological changes in the images are obvious with the change of time.

The other dataset LGC is about the drainage area in the north of new Wales. The height is 80 km in the north–south direction, and the width is 68 km in the east–west direction. The image pairs are Landsat-5 Thematic Mapper (Landsat-5 TM) image and MODIS MOD09GA product image. The image size of Landsat-5 TM is 3200×2720. Similarly, the MODIS data are upsampled to the same size as the Landsat image. There are 14 pairs of images in the LGC dataset. The scenes provided by LGC are mainly the watershed areas. With the change of time, the land coverage changes significantly.

B. Experiments Settings

This article designed four part experiments to validate the proposed method MLFF-GAN. First, we compared MLFF-GAN with five state-of-the-art algorithms (FSDAF [10], BiaSTF [31], enhanced DCSTFN (EDCSTFN) [28], GAN-STFM [36], and SSTSTF [34]) to prove its effectiveness. Second, we show the function of different parts in MLFF-GAN by ablating different parts of its structure in the generator. Third, we also test the scenario of two image inputs for MFLL-GAN and compare them with results by three image inputs. Finally, the computation load and time consumption of network training and prediction are discussed.

In addition to the GAN-STFM algorithm [36], the image pairs near the time are divided into a group. In the CIA dataset, the training data are 11 groups, which are composed of the first ten pairs of images in time, and the test data are the other 5 groups, which are composed of the last six pairs of images in time. Two pairs of images in a group are adjacent in time. Similar to the CIA dataset, in the LGC dataset, the training data are nine groups, and the test data are the other four groups. In the training phase, the image pairs of the later period are allowed to be used as known images, while the low-resolution images of the previous period are allowed to be used as MODIS images of input at prediction time.

For MLFF-GAN generator, because the scale ratio between the Landsat image and the MODIS image is about 16, the network of downsampling feature extraction and upsampling image reconstruction is with four layers. The input of the generator is a six-band image. The numbers of channels for each block in downsampling decode are 16, 32, 64, 128, and 256, respectively. Similar to U-net, the numbers in upsampling are 256, 128, 64, 32, and 16. Also, the output is a six-band image, too.

For MLFF-GAN discriminator, the inputs are the low-resolution image and the high-resolution image, and the total number of input bands is 12. The output is a matrix in the range of 0–1, and the closer the value is to 1, the more realistic the image is.

Four evaluation indexes were used in the experiment: mean absolute error (MAE), root mean squared error (RMSE), spectral angle mapper (SAM) [54], and SSIM index [55]. MAE and RMSE are indicators to evaluate the overall difference between the prediction and the ground truth. RMSE is more sensitive to outliers. SAM is used to evaluate the degree of spectral distortion of the results. The larger the SAM value is, the greater the spectral difference is. SSIM is used to measure the SSIM between the prediction and the ground truth. A high SSIM value indicates that the two images are more similar in a texture structure.

The values of parameters α, β, γ, and δ in (10) will affect the performances of the proposed fusion method. They are two types of loss, the GAN loss and the image content loss. The parameter α is the weight of GAN loss, and it maintains the authenticity of the image, which is specifically reflected in the texture clarity of the image. Without α, the fusion image will be blurring and lack some details. The parameters β, γ, and δ are the weight of the loss for image contents. With β, the fusion image will not only be estimation in a statistical sense globally but also with features corresponding locally. γ will ensure the spectrum similarity between the fusion image and the input images. δ will ensure the similarity texture structure between the fusion image and the input images.

Among the four losses ($\mathcal{L}_{\text{GAN}}(G)$, $\mathcal{L}_1$, $\mathcal{L}_{\text{spectrum}}$, and $\mathcal{L}_{\text{structure}}$), $\mathcal{L}_1$ loss is essential, because it directly calculates the difference between the predicted images and the target images. The other three losses are assistant to make the results better. As these four losses have a certain correlation and $\mathcal{L}_1$ loss is essential, we let $\{\alpha = 1 \times 10^{-1}, 1 \times 10^{-2}, 1 \times 10^{-3}\}$, $\beta = 1$, $\gamma = \{0, 0.5, 1, 1.5, 2\}$, $\delta = \{0, 0.5, 1, 1.5, 2\}$ and carried out 75 groups of experiments to observe the effects of α, γ, and δ on the results. The quantitative index evaluations are shown in Fig. 6. The contributions of GAN loss to the quality of generated images are shown in Fig. 7 by the generated images with different α values.

Comprehensively considering the quantitative indexes and visual results, the parameters in (10) are set as: $\alpha = 0.01$ and $\beta = \gamma = \delta = 1$.

The learning rate for training is set to 2×10^{-4} for BiaSTF, GAN-STFM, SSTSTF, and MLFF-GAN, and 1×10^{-3} for EDCSTFN. In train data, the clipping interval of the image is 200 pixels, and the training algebra of the model is 200 epochs.

C. Experimental Results and Analysis

In Sections III-C1 and C2, these four indexes (MAE, RMSE, SAM, and SSIM) have been used to evaluate the experiment

[1] https://dx.doi.org/10.4225/08/5111AC0BF1229CIA
[2] https://dx.doi.org/10.4225/08/5111AD2B7FEE6LGC

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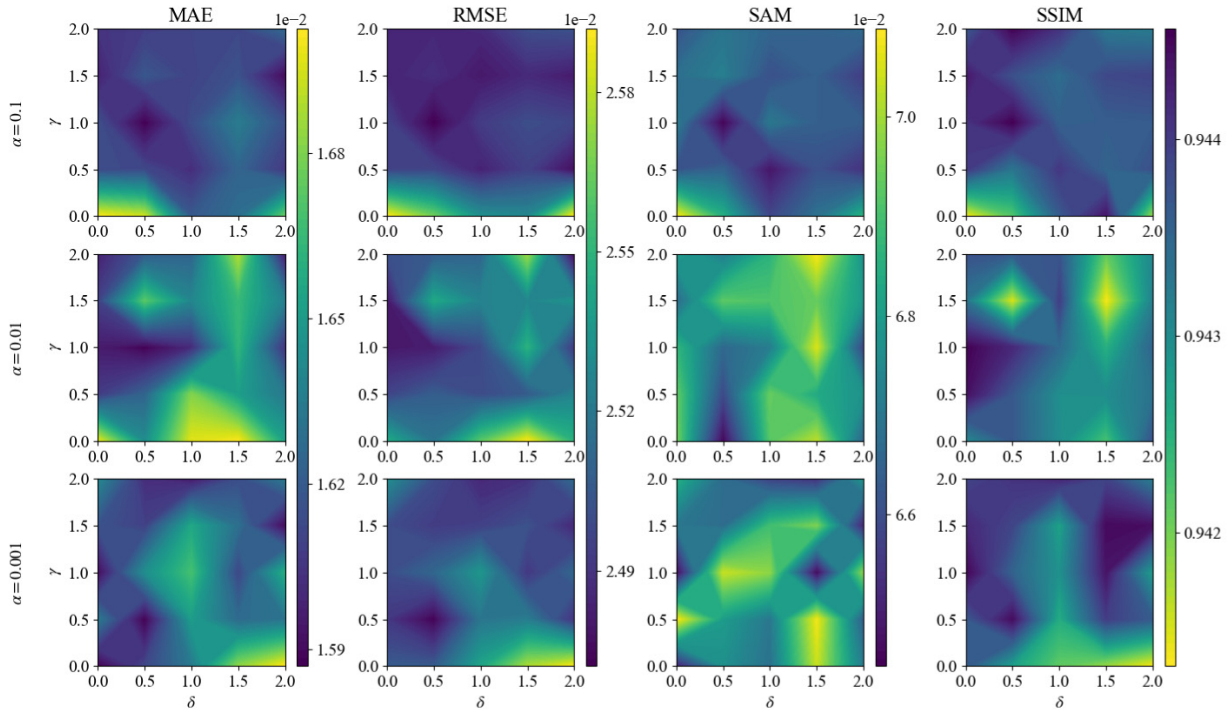


Fig. 6. Comparison experiments on sensitivity analysis by 75 groups of parameters.



Fig. 7. Images generated with different α values, and $\beta = \gamma = \delta = 1$. (a) $\alpha = 0$. (b) $\alpha = 0.001$. (c) $\alpha = 0.01$. (d) $\alpha = 0.1$.

results quantitatively; at the same time, we also validate the proposed method by visual quality comparisons.

1) Contrast Experiments: Five methods, FSDAF, BiaSTF, EDCSTFN, GAN-STFM, and SSTSTF, are selected for comparison. FSDAF is an algorithm based on weight and unmixing, which shows good performance as a non-deep learning method. BiaSTF, EDCSTFN, GAN-STFM, and SSTSTF are methods based on deep learning. BiaSTF and EDCSTFN use the architecture of a CNN. GAN-STFM uses the architecture of GANs. SSTSTF is a multistage network, and it consists of two CNN networks and a GAN network. For FSDAF, EDCSTFN, SSTSTF, and MLFF-GAN, the input data are a low-resolution image at the prediction time and a pair of recent low-high resolution images. For BiaSTF, one more pair of reference images as inputs. For GAN-STFM, the input data are a low-resolution image at the prediction time and a recent high-resolution image. It should be noted that the SSTSTF method is slightly different from that in the original paper. To compare with our experiment, the four-band image in the SSTSTF of its original paper is changed to a six-band image. Besides, we notice that the GAN-STFM algorithm only needs two input images.

Tables I and II show the quantitative index evaluation of the six algorithms on the CIA dataset and the LGC dataset, respectively.

For the CIA dataset as in Table I, in most cases, the proposed MLFF-GAN shows better performances.

However, the FSDAF algorithm also shows some good performances, and in some cases, its quantitative index overpasses some of deep learning methods. We think that there are two reasons: first, the features in the CIA dataset are obvious, so that the pixels are easy to be classified into their right categories. For the FSDAF method, the categories are their end-member in spectral unmixing, so the CIA dataset is more friendly to FSDAF. Second, the data volume of the CIA dataset is relatively small, so that the deep learning algorithm is easy to overfit. Furthermore, a small amount of data are not conducive to learning spatial–temporal features by deep architecture as EDCSTFN or GAN-STFM.

The LGC dataset has a larger volume than the CIA data, so in Table II, the deep learning algorithms (EDCSTFN, GAN-STFM, and MFLL-GAN) show better performances than FSDAF. SSTSTF performs poorly in the LGC dataset. We believe that the uncertainty of upsampling method leads

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TABLE I
RESULTS OF CONTRAST EXPERIMENT ON CIA DATASET

Date	Model	MAE	RMSE	SAM	SSIM
Apr. 1st	FSDAF	0.01996	0.0297	0.08182	0.93694
	BiaSTF	0.02117	0.03032	0.0785	0.94014
	EDCSTFN	0.01972	0.029	0.08232	0.94226
	GAN-STFM	0.02332	0.03305	0.10131	0.92563
	SSTSTF	0.01975	0.02953	0.08116	0.94186
	ours	**0.01898**	**0.02781**	**0.07647**	**0.94453**
Apr. 10th	FSDAF	0.02054	0.02974	0.07967	0.93381
	BiaSTF	0.02	0.02834	0.07299	0.94488
	EDCSTFN	0.02034	0.02901	0.07476	0.94265
	GAN-STFM	0.02177	0.03186	0.09831	0.92507
	SSTSTF	0.02005	0.02939	0.07682	0.9433
	ours	**0.01627**	**0.02454**	**0.06657**	**0.95023**
Apr. 17th	FSDAF	0.01712	0.02684	0.08199	0.92987
	BiaSTF	0.01917	0.02849	0.08651	0.93071
	EDCSTFN	0.01877	0.02755	0.08016	0.92746
	GAN-STFM	0.02205	0.03302	0.11265	0.91024
	SSTSTF	**0.01689**	**0.0261**	0.08201	0.93246
	ours	0.01732	0.02645	**0.07726**	**0.93297**
Apr. 26th	FSDAF	0.01743	0.02731	0.07102	0.92604
	BiaSTF	0.01767	0.02725	0.07312	0.93147
	EDCSTFN	0.01727	0.02677	0.06504	0.93386
	GAN-STFM	0.02133	0.03293	0.09565	0.91319
	SSTSTF	0.01735	0.02733	0.07312	0.93147
	ours	**0.01682**	**0.02656**	**0.06462**	**0.93506**
May. 3rd	FSDAF	0.01494	0.02376	0.06009	0.94411
	BiaSTF	\	\	\	\
	EDCSTFN	0.01534	0.02387	0.05616	0.94794
	GAN-STFM	0.01869	0.02945	0.0828	0.92959
	SSTSTF	0.01544	0.02424	0.06251	0.94769
	ours	**0.0146**	**0.02304**	**0.05316**	**0.9495**
Mean	FSDAF	0.018	0.02747	0.07492	0.93415
	EDCSTFN	0.01829	0.02724	0.07169	0.93883
	GAN-STFM	0.02143	0.03206	0.09814	0.92074
	SSTSTF	0.0179	0.02732	0.07512	0.93936
	ours	**0.0168**	**0.02568**	**0.06762**	**0.94246**
	BiaSTF	0.0195	0.02835	0.07758	0.9368
	ours*	**0.01735**	**0.02434**	**0.07123**	**0.9407**

* Since BiaSTF uses two reference time image pairs to predict the results, only the average value of the first 4 days is calculated here.

TABLE II
RESULTS OF CONTRAST EXPERIMENT ON LGC DATASET

Date	Model	MAE	RMSE	SAM	SSIM
Jan. 29th	FSDAF	0.01849	0.02822	0.07604	0.9457
	BiaSTF	0.0243	0.03573	0.09619	0.94154
	EDCSTFN	0.01739	0.02637	0.07396	0.95082
	GAN-STFM	0.01805	0.02669	0.08504	0.94651
	SSTSTF	0.01946	0.02838	0.08357	0.94966
	ours	**0.01599**	**0.02434**	**0.06878**	**0.95599**
Feb. 14th	FSDAF	0.01573	0.02407	0.06086	0.96013
	BiaSTF	0.02336	0.03356	0.079	0.96948
	EDCSTFN	0.01448	0.02143	0.05963	0.96841
	GAN-STFM	0.01473	0.02201	0.06721	0.96464
	SSTSTF	0.01509	0.02236	0.06883	0.96394
	ours	**0.01277**	**0.01909**	**0.05458**	**0.97129**
Mar. 2nd	FSDAF	0.01214	0.0183	**0.0491**	0.97322
	BiaSTF	0.01417	0.02236	0.0502	0.97481
	EDCSTFN	0.01287	0.01894	0.05453	0.97296
	GAN-STFM	0.01345	0.01976	0.06326	0.9711
	SSTSTF	0.01493	0.02126	0.06236	0.97214
	ours	**0.01165**	**0.01718**	0.04944	**0.97627**
Apr. 3rd	FSDAF	0.01451	0.02151	0.05812	0.9665
	BiaSTF	\	\	\	\
	EDCSTFN	0.01512	0.02242	0.06116	0.96823
	GAN-STFM	0.01493	0.02127	0.06101	0.96911
	SSTSTF	0.01598	0.02298	0.06514	0.9693
	ours	**0.01427**	**0.02007**	**0.05257**	**0.97243**
Mean	FSDAF	0.01522	0.02302	0.06103	0.96139
	EDCSTFN	0.01496	0.02229	0.06232	0.96511
	GAN-STFM	0.01529	0.02243	0.06913	0.96284
	SSTSTF	0.01636	0.02376	0.06997	0.96376
	ours	**0.01367**	**0.02017**	**0.05634**	**0.969**
	BiaSTF	0.02061	0.03055	0.07513	0.96194
	ours*	**0.01347**	**0.0202**	**0.0576**	**0.96786**

* Since BiaSTF uses two reference time image pairs to predict the results, only the average value of the first 3 days is calculated here.

to this result. BiaSTF performs poorly in both datasets. In the deep learning methods, a network architecture needs to be more suitable for spatiotemporal fusion to have better performance. Spatiotemporal fusion as an ill-posed problem is difficult to achieve good performance only using model-driven schemes and handcraft features. In the case of large training data, deep learning with plenty of parameters can find the mapping relationship between the different resolution of images, so as to reduce the prediction error of the empirical model. Relatively speaking, the overall performances of the deep learning method are better than the non-deep learning method, especially in Table II.

In Tables I and II, compared with other methods, the most obvious advantage of MFLL-GAN is the SAM index. For SSIM index, the advantage of MFLL-GAN is very slight. The performances of GAN-STFM often have no large advantage partly, because it only used two input images.

Figs. 8 and 9 are visual comparisons for different methods. The design of AdaIN and AM allows the proposed MFLL-GAN to detect changes in different areas from both local and global features. Figs. 8 and 9 (top) show the global images of results by false color (bands 4, 3, and 2) from different algorithms, and Figs. 8(a) and 9(a) (bottom) show the real Landsat image at the predicting time. Figs. 8 and 9 (middle) are zoomed areas within the white frame in global images. Figs. 8(b)–(g) and 9(b)–(g) (bottom) are the sum of the absolute value difference of all six bands between the ground truth and the predicted results about different methods. Figs. 8(a) and 9(a) are the difference maps between the real image at the reference time and the real image at the current time. In Figs. 8 and 9, please note that Figs. 8(a) and 9(a) (bottom) are the difference between t_1 and t_2, but Figs. 8(b)–(g) and 9(b)–(g) (bottom)

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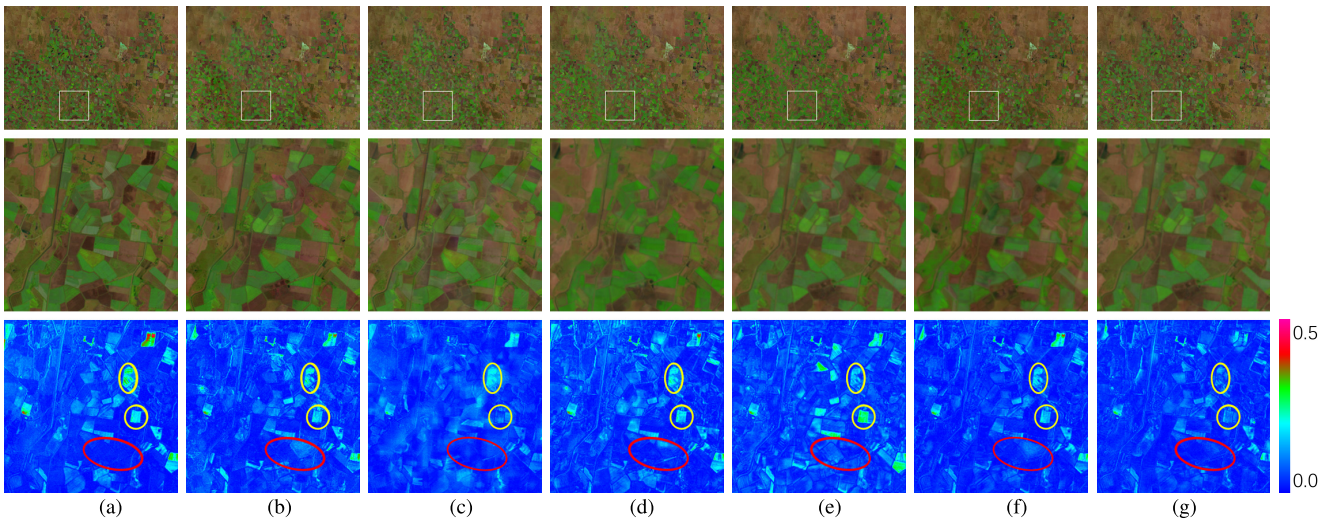


Fig. 8. Results of six algorithms on the CIA dataset on April 10. Middle images are the enlarged images of the white box area in top images. Bottom (a) is the absolute value difference map between the real high-resolution images at t_1 and t_2. Bottom (b)–(g) are the absolute value difference map between the real image. Please note that bottom (a) is the difference between t_1 and t_2, but bottom (b)–(g) are difference between the real image and the prediction, both at t_2.

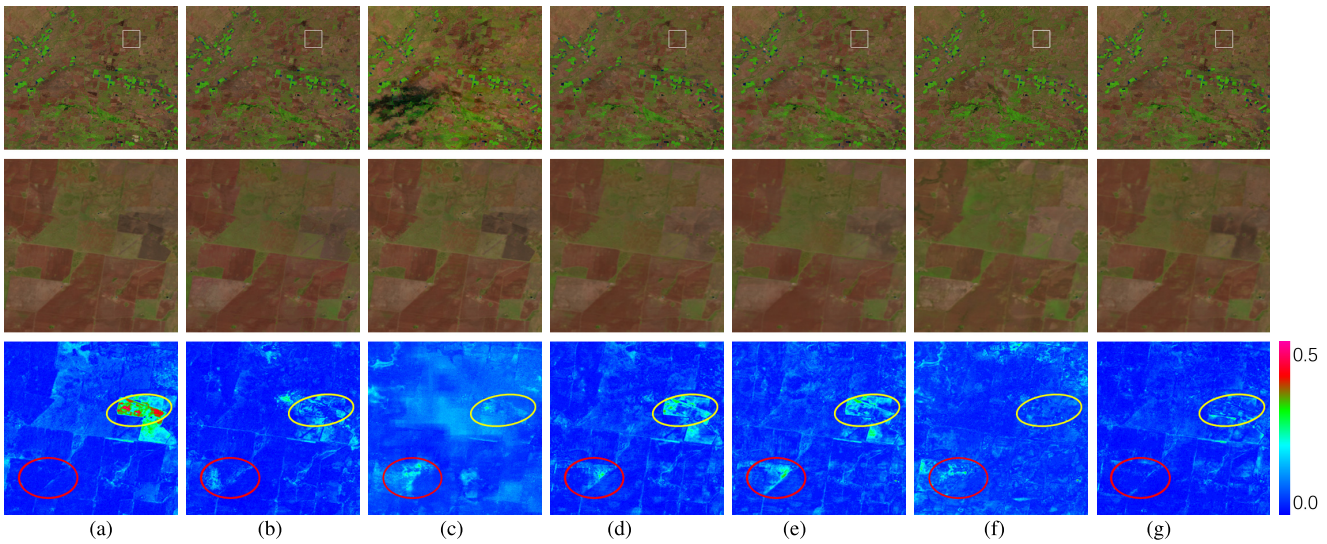


Fig. 9. Results of six algorithms on the LGC dataset on April 10. Middle images are the enlarged images of the white box area in top images. Bottom (a) is the absolute value difference map between the real high-resolution images at t_1 and t_2. Bottom (b)–(g) are the absolute value difference map between the real image. Please note that bottom (a) is the difference between t_1 and t_2, but bottom (b)–(g) are difference between the real image and the prediction, both at t_2.

are the difference between the real and predicted images, all at t_2.

a) Fusion for local abrupt features benefits from the design of AM module: AM generates a weight map by extracting and comparing features of the MODIS image at the reference time and the prediction time. The weight is used to determine the intensity when injecting the features of the high-resolution image at reference time or the features of the low-resolution image at the prediction time. For the small areas with strong local change, the feature weight of the high-resolution image at the reference time is larger. In this way, it keeps the fusion feature consistent with that of the high-resolution image at the reference time in the area without obvious change. For example, in bottom Fig. 8(a), the red circle in the image is dark blue, so it means that the change is very small. For the FSDAF algorithm shown in Fig. 8(b) (bottom), the classification process for end-members cannot adaptively change with dynamic features, so the deviation in these areas will be very obvious when the unchanged pixels were classified into wrong classes. We can observe that there are also large errors in these areas by the deep learning methods in Fig. 8(c)–(f) (bottom). Because the deep learning method may rely too much on the learned experiences, Fig. 8(c)–(f) (bottom) has more obvious prediction errors in the red circle areas. The MLFF-GAN performs best in these areas, because its AM reduces misjudgment on unchanged features. The parts in the area with a red circle show obvious prediction error almost by all methods. Although for MLFF-GAN in Fig. 8(g), the red circle area also predicts the wrong changes, and the wrong prediction is greatly weakened due to the constraints of the AM module. For GAN-STFM, because only two images are taken as input, the network will allocate the

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information of low-resolution image at the prediction time into the high-resolution image at the reference time to a certain extent. It is easy to retain the error caused by different sensors, so that as in Fig. 8(e), the red circle areas with small differences from the previous image will have greater prediction errors. Similarly, on the LGC dataset, the red circle areas of Fig. 9(b)–(f) also show the same results as the CIA dataset. MLFF-GAN shows better results where there is little difference between the prediction time and the reference time.

For the part with a large local difference, the weight of high-resolution feature of the reference time is smaller, while the weight of low-resolution feature is larger, so that the fusion features have more low-resolution spectral information of prediction time in the area with larger change. In Fig. 9(a) (yellow circle areas) in the LGC dataset, we can see that the MLFF-GAN in Fig. 9(g) produces better results in a large range compared with Fig. 9(c)–(f). Similarly, in Fig. 8(a) (two yellow circles) in the CIA dataset, compared with Fig. 8(b)–(f), Fig. 8(g) shows fewer prediction errors for the parts with larger differences.

b) Reconstruction for global spectral textures benefits from AdaIN: It keeps the MLFs of high-resolution images consistent with those of low resolution images in overall distribution (such as mean and variance). In Fig. 8(b)–(g) (left half part of bottom) in the CIA dataset, we can clearly see that MLFF-GAN is more similar to real images in a global distribution compared with other algorithms. In addition, MLFF-GAN also performs better in the prediction error map of Fig. 8(g). In the case of large-scale changes in images, such as LGC, the advantage of AdaIN is more obvious. We can see that in Fig. 9(g) (left half part), compared with Fig. 9(b)–(f), the overall prediction error is less, which is basically similar to that of real images.

2) Ablation Study: There were four ablation experiments: 1) deleting copy and crop in the U-Net architecture (CAC), the two features are fused in the last layer of the decoder, and then upsampled until the image is generated; 2) the strategy of CAC is adopted, but the AM and AdaIN are not added, and only the MLFs are averaged and transferred; 3) CAC is adopted, and AM is added; and 4) CAC, AM, and AdaIN are all added. Figs. 10 and 11 are the quality comparisons in visual, and Table III and IV are the quantitative comparisons.

Table III shows the quantitative experimental evaluation of ablation experiments on the CIA dataset. For the CIA dataset, in terms of quantitative evaluation, the performance is the worst when only using CNN without CAC, AM, and AdaIN. Only using CNN, the features lose too much information in the downsampling, so it is difficult to reconstruct the image details. Although the spectral information of the low-resolution image is fused at the bottom, without the help of CAC, it brings too much uncertainty to the spectral information of the whole image. In Tables III and IV, after adding CAC, the network can capture the previous details and can generate higher quality images, which greatly improves the fusion performances for all dates of experiments.

In Table III, when the AM module is added to determine the weight in using different resolutions of image features, it will improve the fusion performance in most cases. Adding the AdaIN module can also improve the performance by ensuring the global statistical consistency between the fusion results and the ground truth. From the quantitative evaluation,

TABLE III
ABLATION EXPERIMENT ON CIA DATASET

Date	Model			MAE	RMSE	SAM	SSIM
	CAC	AM	AdaIN				
Apr. 1st	×	×	×	0.0214	0.0307	0.08933	0.92649
	√	×	×	0.01971	0.02911	0.07936	0.9423
	√	√	×	0.01937	0.02844	0.07841	0.94393
	√	√	√	**0.01898**	**0.02781**	**0.07647**	**0.94453**
Apr. 10th	×	×	×	0.01797	0.0265	0.07604	0.9352
	√	×	×	0.01786	0.02643	0.07036	0.94589
	√	√	×	0.01771	0.02611	0.07013	0.94861
	√	√	√	**0.01627**	**0.02454**	**0.06657**	**0.95023**
Apr. 17th	×	×	×	0.01885	0.0272	0.08954	0.92214
	√	×	×	0.01861	0.02783	0.0838	0.92667
	√	√	×	**0.01722**	**0.02627**	**0.076**	**0.93323**
	√	√	√	0.01732	0.02645	0.07726	0.93297
Apr. 26th	×	×	×	0.01868	0.02802	0.07634	0.92447
	√	×	×	0.01723	0.02747	0.06773	0.93169
	√	√	×	**0.01663**	**0.02615**	0.06491	**0.93512**
	√	√	√	0.01682	0.02656	**0.06462**	0.93506
May. 3rd	×	×	×	0.0165	0.02502	0.0674	0.93409
	√	×	×	0.01556	0.02428	0.05616	0.94629
	√	√	×	0.01563	0.02394	0.05473	0.94897
	√	√	√	**0.0146**	**0.02304**	**0.05316**	**0.9495**
Mean	×	×	×	0.01868	0.02749	0.07973	0.92848
	√	×	×	0.01779	0.02702	0.07148	0.93857
	√	√	×	0.01731	0.02618	0.06884	0.94197
	√	√	√	**0.0168**	**0.02568**	**0.06762**	**0.94246**

TABLE IV
ABLATION EXPERIMENT ON LGC DATASET

Date	Model			MAE	RMSE	SAM	SSIM
	CAC	AM	AdaIN				
Jan. 29th	×	×	×	0.02013	0.03002	0.08836	0.93552
	√	×	×	0.01726	0.02598	0.0698	0.95459
	√	√	×	0.01797	0.02669	0.06991	0.95479
	√	√	√	**0.01599**	**0.02434**	**0.06878**	**0.95599**
Feb. 14nd	×	×	×	0.02248	0.03339	0.08484	0.9385
	√	×	×	0.01265	0.01889	0.05706	0.97049
	√	√	×	**0.01245**	**0.01865**	0.05601	0.96984
	√	√	√	0.01277	0.01909	**0.05458**	**0.97129**
Mar. 2nd	×	×	×	0.01892	0.02765	0.07771	0.94973
	√	×	×	0.01343	0.01899	0.05011	0.97479
	√	√	×	0.01391	0.01959	0.05037	0.97421
	√	√	√	**0.01165**	**0.01718**	**0.04944**	**0.97627**
Apr. 3rd	×	×	×	0.01898	0.02691	0.07566	0.94697
	√	×	×	0.01799	0.02419	0.05415	0.96799
	√	√	×	0.01793	0.02412	0.05266	0.96877
	√	√	√	**0.01427**	**0.02007**	**0.05257**	**0.97243**
Mean	×	×	×	0.02013	0.02949	0.08164	0.94268
	√	×	×	0.01533	0.02201	0.05778	0.96696
	√	√	×	0.01556	0.02226	0.05724	0.96690
	√	√	√	**0.01367**	**0.02017**	**0.05634**	**0.969**

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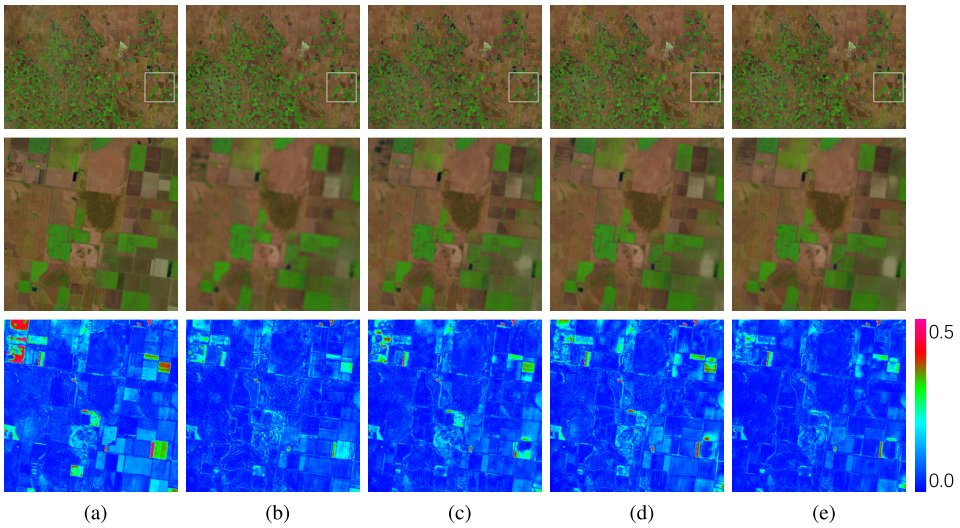


Fig. 10. Results of ablation experiments on the CIA dataset on April 26. (a) Real image. (b) Without CAC, AM, and AdaIN. (c) With CAC, without AM and AdaIN. (d) With CAC and AM, without AdaIN. (e) With CAC, AM, and AdaIN. Middle images are the enlarged images of the white box area in the top images. Bottom (a) is the absolute value difference map between the real high-resolution images at t_1 and t_2. Bottom (b)–(e) are the difference maps between the real image and the prediction, both at t_2.

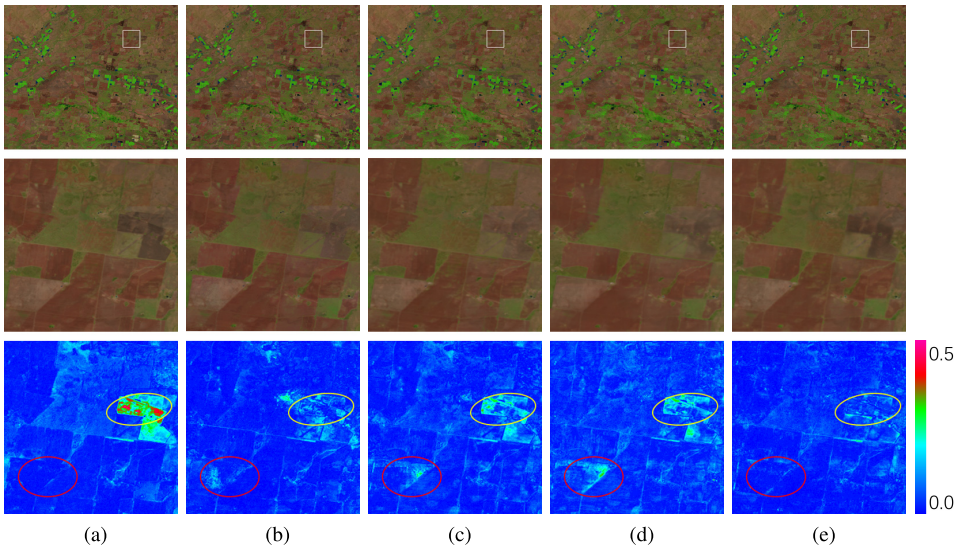


Fig. 11. Results of ablation experiments on the LGC dataset on April 26. (a) Real image. (b) Without CAC, AM, and AdaIN. (c) With CAC, without AM and AdaIN. (d) With CAC and AM, without AdaIN. (e) With CAC, AM, and AdaIN. Middle images are the enlarged images of the white box area in the top images. Bottom (a) is the absolute value difference map between the real high-resolution images at t_1 and t_2. Bottom (b)–(e) are the difference maps between the real image and the prediction, both at t_2.

we can observe that, on the whole, the generator with AM and AdaIN performs better. However, in the two images on April 17 and 26, AdaIN does not show obvious effectiveness. After checking the images, we find that, in these two periods, the large-scale changes are relatively less, so the advantages of AdaIN are hard to play out. On the contrary, only the AM module for local features has the chance to perform better. From the performance of SAM index on April 26, it can be seen that the AdaIN module has a more positive effect on the overall performance.

To further observe the role of attention maps in the network, we output them in the experiments of the CIA dataset on April 10, as shown in Fig. 12. It can be seen that the weight of attention is not directly corresponding to the image difference. Formula (4) also means that the features from different layers need different weight maps of attention. Compared with others, M_0 is smoother and has less noise, less receptive fields, and richer weight information. In M_0 scale, in the area where C_2 cannot reflect texture information, especially the farmland part, M_0 basically gives all the weight to $\hat{A}_i^{F_1}$ at this scale. In M_1 scale, although the AM can provide a small amount of weights in the figure, most of them are noise, and the weights are basically maintained around 0.3. It means that the network gives F_1 more weights to select more spatial information to ensure the quality of the generated image, while C_2 only accounts for a small part of the weights to conserve the spectral

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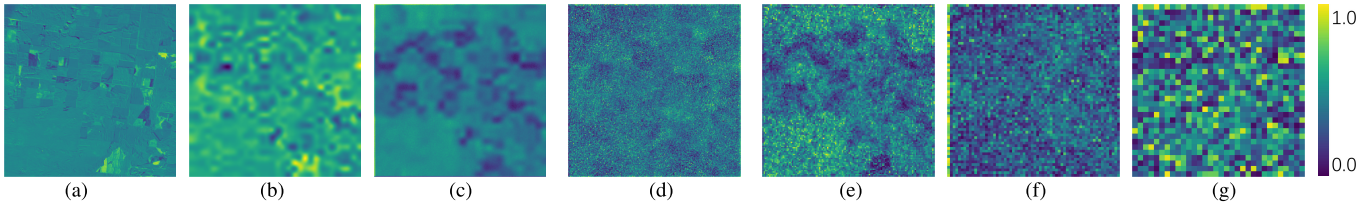


Fig. 12. Difference map of images and the weight maps of attentions. (a) $|F_1 - F_2|$. (b) $|C_1 - C_2|$. (c) M_0. (d) M_1. (e) M_2. (f) M_3. (g) M_4.

quality of the generated result. In M_2 scale, there is still a lot of noise, but from the overall image, with weak textures has a significantly higher weight on C_2 than other parts. In M_3 and M_4 scales, they usually do not provide guidance information with interpretability under CIA dataset, and most of them are residual compensations and look like noise.

For the LGC dataset in Table IV, because it is flood-related data and has a wide range of spectral information changes, the AdaIN module shows a stronger advantage. The CIA dataset in Table III includes many local changes of block farmland, so the AM block will be more useful.

Figs. 10 and 11 show the predicted results of each ablation experiment on CIA set on April 10 and LGC set on November 25. In the visual comparison, Figs. 10(b) and 11(b) (middle) show the image generated without CAC. Both of them are very fuzzy and lack details. In Fig. 10(b) (bottom), the prediction image shows less extreme values in a wide range. Also, in Fig. 11(b) (bottom), the prediction image shows errors in a wide range. Because the model without CAC only fuses the features in the last layer of the decoder and lacks the information of multilayer structure, inaccurate prediction is often caused in a large range in the reconstruction process. In Figs. 10 and 11, please note that Figs. 10(a) and 11(a) (bottom) are the difference between t_1 and t_2, but Figs. 10(b)–(e) and 11(b)–(e) (bottom) are the difference between the real and predicted images, all at t_2

Comparing Fig. 10(c)–(e) (bottom), the difference of some farmland blocks becomes smaller in the result of the model with AM and AdaIN. However, large deviation in areas with a size of 2–3 pixels (the red areas) cannot be removed. Because the difference weight is guided by two low resolution images at different times, it is not very easy to find the small-scale change between the images. However, the input of the generator is two low-resolution bicubic interpolation images, so it is more effective for the large-scale change areas.

Comparing Fig. 10(d) and (e) (bottom) with Fig. 11(d) and (e) (bottom), AdaIN makes the whole error map more "monotonous." For the absolute value difference map, it is usually good if there are more dark blue areas and fewer light yellow areas. After adding AdaIN transformation of high-resolution features, the overall statistical characteristics of the fusion images are more similar to that of real images at the prediction time, which means that the overall difference is smaller.

The MLFs are also important improvements for U-net in the feature extraction and image reconstruction stage. Except the four ablation experiments, to prove the effectiveness of MLF, we add an ablation experiment to remove the multilevel structure. To compare the computing consumption, we calculated the parameters and multiply–accumulate operations [56], [57] (MACs) using an image of 256 × 256 with six bands) for the two situations, as shown in Table V. To compare the network performances, the quantitative evaluation of the two networks is shown in Table VI.

TABLE V
PARAMETERS AND MACS OF GENERATOR WITH MLF AND WITHOUT MLF

	Parameters	MACs
With MLF	5.7×10^5	1.3×10^{10}
Without MLF	5.1×10^6	4.6×10^{11}

TABLE VI
ABLATION EXPERIMENT OF MLF

	MAE	RMSE	SAM	SSIM
With MLF	0.01685	0.0256	0.06959	0.94048
Without MLF	0.01851	0.02777	0.0735	0.94046

In terms of network parameters and computing speed in Table V, the fusion with MLFs shown large advantages compared with the fusion without MLFs, because the size of intermediate features is greatly reduced and the receptive field with multilevel structures can be better suitable for feature fusion of different temporal and spatial resolutions. The fusion performances are also obviously improved with MLFs, as shown in Table VI.

The most important function of GAN loss is to improve the quality of the generated image and makes the image less blurred. This function is proven by the ablation of GAN loss. As shown in Fig. 7, the generated image with optimal weights has better visual expression. Also, we noticed that a component of a fusion evaluation index proposed by Piella and Heijmans [58] can evaluate the edge integrity and sharpness of the fused image results. The component (e-SSIM) calculates the SSIM between the edge detection results of the real image and the fusion image. As shown in Tables VII and VIII, the generated images with GAN have obvious advantages on texture clarity. Besides, for other four indexes, the results with GAN loss only have certain advantages in SAM, whether to adopt GAN loss has no obvious advantages and disadvantages in MAE, RMSE, and SSIM.

3) Computation Load: To evaluate the computation load, we calculate the parameters and MACs of the deep

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TABLE VII
ABLATION EXPERIMENT OF GAN ON CIA DATASET

	e-SSIM	MAE	RMSE	SAM	SSIM
With GAN	**0.84478**	0.0168	0.02568	**0.06762**	0.94246
Without GAN	0.83248	**0.01661**	**0.02527**	0.06817	**0.94357**

TABLE VIII
ABLATION EXPERIMENT OF GAN ON LGC DATASET

	e-SSIM	MAE	RMSE	SAM	SSIM
With GAN	**0.90354**	0.01367	0.02017	**0.05634**	0.969
Without GAN	0.87609	**0.01363**	**0.02003**	0.05991	**0.96944**

TABLE IX
PARAMETERS AND MACS OF FIVE DEEP LEARNING METHODS

	Parameters	MACs
BiaSTF	3.4×10^6	22.2×10^{10}
EDCSTFN	2.8×10^5	1.8×10^{10}
GAN-STFM	5.8×10^5(Generater)	3.8×10^{10}
	3.6×10^6(Discriminator)	7.7×10^6
SSTSTF	1.1×10^7	12.5×10^{10}
MLFF-GAN	5.9×10^6(Generater)	1.4×10^{10}
	2.8×10^6(Discriminator)	3.8×10^9

learning methods, respectively. MACs is the number of multiply–accumulate operations required by the network to calculate a 256 × 256 image with six bands. The results are shown in Table IX. It should be noted that the BiaSTF shows the computation load by six times, because BiaSTF is a single-band algorithm. The computation load of SSTSTF is a summarization of all the stages of the networks.

Among the five deep learning algorithms, BiaSTF and SSTSTF are multistage deep learning methods. They will lead to more parameters and computation. Among them, the super-resolution networks used in SSTSTF need 4.8×10^6 parameters and 5.1×10^{10} MACs. The convenience of the model is very ignored. Benefit from the multilevel network, even if MLFF-GAN has basically the same quantity of parameters as GAN-STFM, or even more parameters than EDCSTFN, the number of calculations is much less than them. MLFF-GAN has certain advantages in testing speed.

4) Comparison of Two Input and Three Input Images: In GAN-STFM, the model only needs to input two images to predict the high-resolution image of prediction time. We simplify the fusion process in MLFF-GAN and design a prediction mode of two input images. The new fusion process will replace the structure in Fig. 4(c) and (d). The fusion mode of two input images is shown in Fig. 13.

In the fusion mode of two input images, the attention weight is not determined by the features of two low-resolution fusion images, but by the features of the low-resolution image at the prediction time and the high-resolution image at the reference time. The quantitative evaluation of results is shown

TABLE X
RESULTS OF DIFFERENT INPUT MODELS ON CIA DATASET

Model	MAE	RMSE	SAM	SSIM
2-inputs MLFF-GAN	0.01774	0.02682	0.07054	0.93958
3-inputs MLFF-GAN	**0.0168**	**0.02568**	**0.06762**	**0.94246**

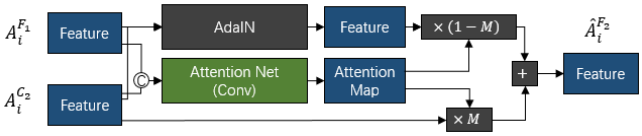


Fig. 13. Simplified model with two input images.

in Table X, which are the average values of all testing set for different dates.

In the mode with three inputs, the features of the low-resolution image are all from a feature extraction network, which has the same level of spatial features and can better make a comparison on the different regions. On the whole, the prediction model with three inputs performs better than the model with two inputs.

IV. CONCLUSION

In this article, we proposed a new spatiotemporal fusion method based on GANs. Three difficulty points are comprehensively considered in the proposed MLFF-GAN, whose generator consists of three stages: feature extraction, feature fusion, and image reconstruction. The backbone of the generator is designed as a U-net-like structure with the encoder–decoder and MLFs. In the fusion stage, AdaIN was introduced to optimize the global spectral distribution, and the AM was used to learn the weights for the local change information. Both quantity and quality comparisons were conducted for the proposed MFLL-GAN and other methods such as FSDAF, BiaSTF, EDCSTFN, GAN-STFM, and SSTSTF. At the same time, the effectiveness of CAC, AdaIN, and AM was evaluated in the ablation experiment. For large resolution gap problem, CAC shows better performances than traditional CNN. With AdaIN and AM, MFLL-GAN shows better performances in both abrupt change areas and global spectrum and texture characteristics. The computation load of different methods was compared and analyzed. We also tested a simplified two input model version for more flexible prediction. We find that three input model has more advantages than the two input model because of utilizing more data and information. Overall, both the contrast and ablation experiments confirmed the advantages of MFLL-GAN when compared with the other five methods on CIA dataset and LGC dataset.

The huge difference in spatiotemporal resolutions is still a challenging problem in the study of spatiotemporal fusion. We find that most deep learning-based methods usually show better performances on CIA dataset than on LGC dataset. The reason may be that the volume of CIA is larger than that of LGC. How to reasonably use more data to solve the

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contradiction between the spatial domain and the temporal domain is the key to the research. In the feature, it is a very promising research direction for some new deep architectures that can better utilize more historical data to alleviate the disturbance of season or phenology.

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